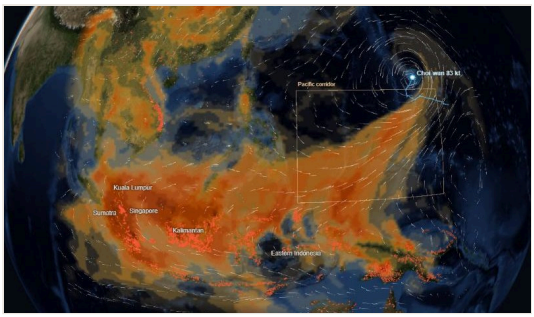
Haze at the typhoon's door: how far the 2026 Indonesian fire smoke reached, in open data



The map globe on 3 October 2026, 07:30 UTC. Smoke near the ground (orange) covers the western islands; a streamer runs north-east from eastern Indonesia, through the Pacific corridor box, into Choi-wan's circulation. Red dots are that day's fires.

On 3 October 2026 a trail of smoke from the Indonesian fires could be seen curling into Super Typhoon Choi-wan, far out over the Pacific. Two animated globes and thirteen Python notebooks follow the haze from the fires to the zone where typhoons form and travel. How unusual was 2026, what carried the smoke that far, and who breathed it on the way?

Lik Ren Tai, November 2026

Open the map globe (also on phones)
(https://likrentai.github.io/hazetyphoon/)

Open the 3D globe (desktop)
(https://likrentai.github.io/hazetyphoon/cesium.html)

Code, notebooks and data notes on GitHub
(https://github.com/likrentai/hazetyphoon)

The question

Almost every dry season, fires on peatland and farmland in Sumatra and Kalimantan send haze over Indonesia, Malaysia and Singapore. 2026 was a strong El Niño year, and the fires were widespread, in eastern Indonesia as well as in the west. On 3 October the global wind and air quality map earth.nullschool.net showed something less familiar: a long trail of haze drawn north-east from eastern Indonesia into Super Typhoon Choi-wan, far out over the Philippine Sea.

The point of this project is the scale of the haze, not a claim that smoke changes typhoons. I wanted to know whether the trail was really smoke, how often typhoons pull smoke out over the Pacific, whether 2026 was unusual, whether the storms also change the haze in the west, and how many people breathed it. The answer comes in two parts: the **direct export** of smoke from eastern Indonesia into the Pacific, and the **indirect effect** of the storms on the haze in the west.

The data

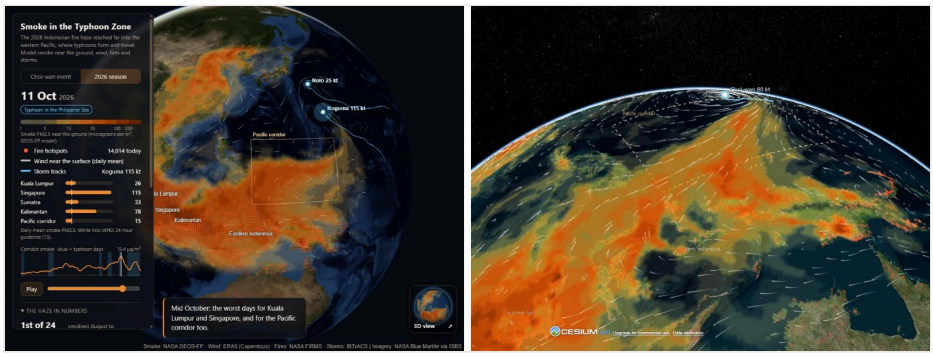
What	Source	Period
Smoke and other aerosols near the ground and in the air column, every 3 hours (0.25 degree) and daily (0.5 degree)	NASA GEOS-FP forecast system, through the NCCS OPeNDAP server	28 Sep to 7 Oct; 1 Aug to 25 Oct 2026

The same, daily, for every season since 2003	NASA MERRA-2 reanalysis (2,166 days, through NASA's cloud OPeNDAP)	Aug to Oct, 2003 to 2026
A second model for the event	Copernicus CAMS (ECMWF)	28 Sep to 7 Oct 2026
Winds near the surface and at 850 and 700 hPa	ERA5 (Copernicus Climate Data Store)	Aug to Oct 2026
Fire hotspots	NASA FIRMS, VIIRS satellites	Aug to Oct 2026
Typhoon tracks and winds	IBTrACS, western Pacific	2003 to 2026
People per square kilometre	WorldPop 2020, 1 km	2020
Ground monitors	Singapore NEA (data.gov.sg); OpenAQ (Kuala Lumpur, Jakarta, Palembang)	Aug to Oct 2026
El Niño and Indian Ocean Dipole indices	NOAA (ONI and DMI)	2003 to 2026

"Smoke" here means the organic, brown and black carbon in the models, not total PM2.5. Around a typhoon most fine particles are sea salt, and a total PM2.5 map, like the nullschool picture, mixes the two. A **typhoon day** is a day with a storm of at least 64 knots in the Philippine Sea box (120 to 150 E, 5 to 25 N), and the **Pacific corridor** is the open ocean between eastern Indonesia and that box (128 to 150 E, 3 to 18 N).

Two globes, thirteen notebooks

Page	What it shows
Map globe (MapLibre and deck.gl)	Smoke, wind, fires and storm tracks, with a switch between the Choi-wan event (3-hourly) and the whole 2026 season (daily), smoke levels in Kuala Lumpur, Singapore, Sumatra, Kalimantan and the corridor, and a small chart with typhoon days. Works on phones.
3D globe (CesiumJS)	The same on aerial imagery, with camera views along the smoke streamer, following the storm, and over Kuala Lumpur and Singapore. Desktop only.
Notebooks (Python)	00 to 04 the Choi-wan event (maps, air trajectories, smoke budgets); 05 to 08 the 2026 season (typhoon days, people exposed, ground check); 09 to 11 the record since 2003; 12 the files for the globes.



Left: the map globe on 11 October, the smokiest day of the season in the corridor, with Typhoon Koguma to the north. Right: the 3D globe on 3 October, looking north-east along the smoke streamer from eastern Indonesia to Choi-wan on the horizon.

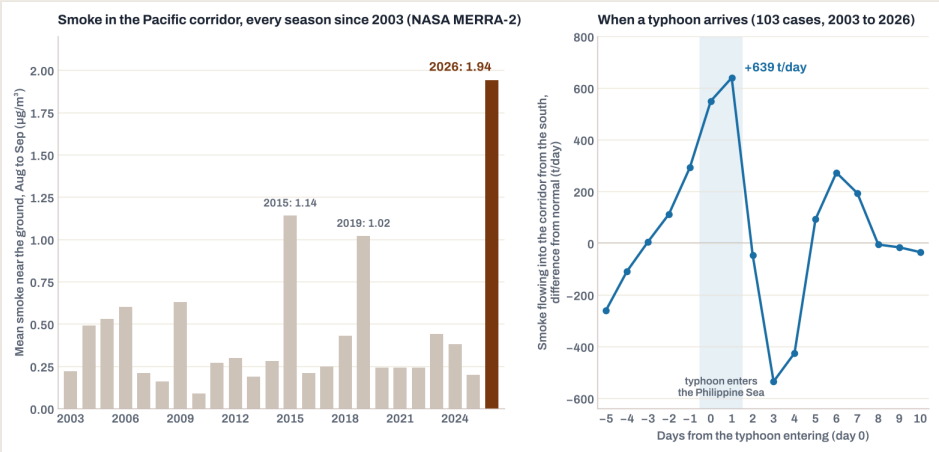
What the data shows

2026 put more smoke in the typhoon zone than any season since 2003

1st of 24

seasons for smoke in the Pacific corridor, August to September (1.94 µg/m³, against 1.14 in 2015)

In the MERRA-2 record, 2026 is the smokiest August and September in both the Pacific corridor and eastern Indonesia (3.81 μg/m^3, against 2.72 in 2015). It is not the record in the west: Kalimantan ranks 3rd, Singapore 4th, Sumatra 5th and Kuala Lumpur 6th, behind 2006, 2015 and 2019. The difference is where the smoke was: in 2026 much more of it sat over eastern Indonesia, upwind of the corridor. The El Niño was the strongest in the period (ONI +2.16 for August to October), and El Niño and the Indian Ocean Dipole together explain 34% to 55% of the year-to-year differences in smoke.



Left: smoke in the Pacific corridor in every season since 2003. Right: smoke flowing into the corridor from the south in the days around a typhoon's arrival, averaged over 103 cases; the shaded days are the arrival.

Typhoons pull smoke out of eastern Indonesia

+29%

more smoke flowing into the corridor from the south on typhoon days, in 24 seasons (+80% in El Niño seasons)

Across all 24 seasons, smoke entering the corridor through its southern side rises by 158 tonnes a day on typhoon days (95% range 29 to 288), and by 430 tonnes a day in the seven El Niño seasons (244 to 613). The smoke in the corridor's air column rises by 28% and 70%. In other seasons the change is small and not clear. Lined up on the day each storm entered the Philippine Sea, the inflow peaks on the day of arrival and the day after, at about 640 tonnes a day above normal, and then drops well below normal as the storm moves on and the winds turn.

The streamer into Choi-wan was real smoke, and it came from the east

2.7M km²

of the Pacific corridor under smoke on 6 October, up from 1.1 million on 29 September

Split into its parts, the trail on the nullschool map is smoke; the bright ring round Choi-wan's eye is sea salt, about 95% of the fine particles near the storm. The edge of the smoke came within about 200 km of the storm's centre. Air parcels released at 850 hPa (about 1.5 km up) over each source region show where it came from: 47% of the parcels from eastern Indonesia reached the corridor and a few (2%) reached the storm itself, after about four and a half days, against 24% from Kalimantan and none from Sumatra. The western fires were simply too far "inside" the continent for the storm's winds to reach.

About half of it was removed on the way

161,800 t

of smoke carried into the corridor over six days of the event; about half was removed before it could leave

A smoke budget for each box (what flows in, what flows out, and the change in what is there) shows eastern Indonesia exporting about 21,000 tonnes of smoke a day during the event, mostly northward, while Sumatra was a net importer. In the corridor, inflow minus outflow minus the change in storage leaves about half of the inflow unaccounted for: the part removed on the way, by rain and settling, plus any error in the budget.

Back home, the storms barely change the haze

-7%

smoke over Sumatra on typhoon days; no clear change in Kalimantan, Kuala Lumpur or Singapore

The indirect effect is weak. On typhoon days the smoke field shifts north and north-east, a little less over Sumatra and southern Kalimantan and a little more over northern Borneo, the southern South China Sea and the corridor, but for Kuala Lumpur and Singapore the change is within the noise.

Tens of millions breathed it

17 to 45 million

people in Indonesia in unhealthy fire smoke on the worst day (above 55.5 µg/m³), depending on the method

Combining the daily smoke with where people live, the worst day put 17 to 45 million Indonesians, 3.7 to 19 million Malaysians and all of Singapore's 4.6 million in unhealthy levels of fire smoke. The range comes from four versions of the calculation: the model as it is, and the model scaled down by the factor found against Singapore's monitors (below), each for fire smoke and for total PM2.5.

Worst day, unhealthy fire smoke	Scaled to monitors	Model as it is
Indonesia	17.1 million	45.0 million
Malaysia	3.7 million	18.6 million
Singapore	4.6 million	4.6 million

Problems in the data, and how they were handled

- **Sea salt that looks like haze.** Near the storm, total PM2.5 is about 95% sea salt. Every finding about the storm uses smoke only (organic, brown and black carbon), and the maps show smoke, not total PM2.5.
- **A model that reads high.** Against Singapore's five official monitoring regions, GEOS-FP tracks the ups and downs well (correlation 0.83 to 0.85) but reads 1.45 to 2.35 times too high, 1.8 times on average. The exposure figures are therefore given as a range, with and without that correction.
- **Few open monitors.** Singapore is the only government with daily open data for the whole season. In Kuala Lumpur, Jakarta and Palembang only low-cost community sensors (AirGradient, through OpenAQ) were available: they agree

with the model in Kuala Lumpur (correlation 0.66 to 0.83) but not in Jakarta, where city pollution dominates, or in Palembang, where the model reads too low near the fires.

- **City pollution counted as smoke.** Organic carbon also comes from traffic and industry, so Jakarta looked smoky on every day. For people's exposure, the project uses brown carbon, which in GEOS-FP comes only from fires.
- **Two models, two levels.** MERRA-2 shows 33% to 68% of GEOS-FP's smoke in 2026, while their daily ups and downs agree (correlation 0.71 to 0.94). Each series is compared only with itself: ranks come from MERRA-2, the event and season from GEOS-FP.
- **A record still being written.** MERRA-2 for October 2026 is not yet published, so the long record covers August and September only and does not include Choi-wan's peak. It will be added in late November.

Limits

All smoke figures come from models that combine satellite data with weather and fire emissions; they are not measurements, and over the open ocean there is nothing on the ground to check them. The link between typhoon days and smoke is an association across 24 seasons, with the season as the unit of uncertainty; it says nothing about whether the smoke changed the storms, and the project does not claim that. The air trajectories cannot resolve the storm's core. Exposure uses 2020 population and smoke near the ground only. The work has not been through peer review; the closest published study, on smoke reaching the Philippines with developing typhoons in 2019 (Reid and others, 2025), looks at single cases rather than the long record.

Where this could be useful

- **Haze monitoring and public health:** which places a season's smoke reaches, how many people it affects, and how much the models need correcting against the few open monitors.
- **Regional forecasting:** a typhoon in the Philippine Sea as an early signal that eastern Indonesian smoke will be carried over the ocean, and washed out on the way.
- **Climate communication and teaching:** a single, striking event checked against two decades of open data, with the uncertainty shown honestly.

About

Lik Ren Tai applies data science in two fields: environmental science and railway systems. This project was built with Python (xarray, pydap, pandas, rasterio, matplotlib), MapLibre GL JS, deck.gl and CesiumJS.

Smoke: NASA GMAO GEOS-FP and MERRA-2; Copernicus Atmosphere Monitoring Service. Winds: ERA5, Copernicus Climate Change Service. Fires: NASA FIRMS (VIIRS). Storms: IBTrACS (NOAA NCEI). Population: WorldPop. Monitors: Singapore NEA (data.gov.sg); OpenAQ and AirGradient. Indices: NOAA. Related work: Reid and others 2025, Atmospheric Chemistry and Physics 25, 18639. Imagery: NASA GIBS; Bing Maps via Cesium ion. Inspired by earth.nullschool.net. This is an independent project, not affiliated with or endorsed by any of the sources.